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Safety Data Sheet according to Regulation (EC) No. 1907/2006 (REACH)

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Trade name/designation:

MiQuant® SP Beads, Binding Buffer

Article No.:

56-4010, 56-4100

UFI:

EEE6-EMSY-Q27T-NR1N

1.2. Relevant identified uses of the substance or mixture and uses advised against

Use of the substance/mixture:

The product is intended for research, analysis and scientific education.

1.3. Details of the supplier of the safety data sheet

Supplier (manufacturer/importer/only representative/downstream user/distributor):

Minerva Biolabs GmbH

Schkopauer Ring 13

12681 Berlin

Germany

Telephone: +49 30 2000437 0

E-mail: ehs@minerva-biolabs.com

Website: www.minerva-biolabs.com

1.4. Emergency telephone number

Vergiftungs-Informations-Zentrale Freiburg, 24h: +49 761 19240

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [CLP]

Hazard classes and hazard categories	Hazard statements	Classification procedure
Acute toxicity (oral) (<i>Acute Tox. 4</i>)	H302: Harmful if swallowed.	Calculation method.
Skin corrosion/irritation (<i>Skin Irrit. 2</i>)	H315: Causes skin irritation.	Calculation method.
Serious eye damage/eye irritation (<i>Eye Irrit. 2</i>)	H319: Causes serious eye irritation.	Calculation method.
Hazardous to the aquatic environment (<i>Aquatic Acute 1</i>)	H400: Very toxic to aquatic life.	Calculation method.
Hazardous to the aquatic environment (<i>Aquatic Chronic 3</i>)	H412: Harmful to aquatic life with long lasting effects.	Calculation method.

2.2. Label elements

Labelling according to Regulation (EC) No. 1272/2008 [CLP]

Hazard pictograms:



GHS07

Exclamation mark



GHS09

Environment

Signal word: Warning

Hazard statements for health hazards	
H302	Harmful if swallowed.
H315	Causes skin irritation.

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Hazard statements for health hazards

H319 Causes serious eye irritation.

Hazard statements for environmental hazards

H400 Very toxic to aquatic life.

H412 Harmful to aquatic life with long lasting effects.

Supplemental hazard information: none

Precautionary statements Prevention

P264 Wash hands thoroughly after handling.

P270 Do not eat, drink or smoke when using this product.

P273 Avoid release to the environment.

P280 Wear protective gloves/protective clothing and eye protection/face protection.

Precautionary statements Response

P301 + P312 IF SWALLOWED: Call a POISON CENTER/doctor/First aider if you feel unwell.

P302 + P352 IF ON SKIN: Wash with plenty of water/Soap.

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P332 + P313 If skin irritation occurs: Get medical advice/attention.

P337 + P313 If eye irritation persists: Get medical advice/attention.

P362 + P364 Take off contaminated clothing and wash it before reuse.

P391 Collect spillage.

Precautionary statements Disposal

P501 Dispose of contents/container to an appropriate recycling or disposal facility.

2.3. Other hazards

No data available

SECTION 3: Composition/information on ingredients

3.2. Mixtures

Hazardous ingredients / Hazardous impurities / Stabilisers:

Product identifiers	Substance name Classification according to Regulation (EC) No 1272/2008 [CLP]	Concentration
CAS No.: 50-01-1 EC No.: 200-002-3 Index No.: 607-148-00-0	guanidinium chloride Acute Tox. 4 (H302), Eye Irrit. 2 (H319), Skin Irrit. 2 (H315)  Warning Acute Toxicity Estimate ATE (oral) 729.8 mg/kg ATE (dermal) > 2,000 mg/kg ATE (inhalation, dust/mist) 3.181 mg/L	25 – ≤ 50 weight-%
CAS No.: 57-09-0 EC No.: 200-311-3	cetrimonium bromide Acute Tox. 4 (H302), Aquatic Acute 1 (H400), Aquatic Chronic 1 (H410), Eye Dam. 1 (H318), STOT RE 2 (H373), STOT SE 3 (H335), Skin Corr. 1 (H314), Skin Irrit. 2 (H315)     Danger M-factor (acute): 100 M-factor (chronic): 1 Specific concentration limit (SCL) Skin Irrit. 2; H315: 2.5% ≤ C < 100% Skin Corr. 1; H314: C ≥ 100% Acute Toxicity Estimate ATE (oral) 410 mg/kg	0.1 – ≤ 2.5 weight-%

Full text of H- and EUH-phrases: see section 16.

SECTION 4: First aid measures

4.1. Description of first aid measures

General information:

In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible). Remove victim out of the danger area. Remove contaminated, saturated clothing. If unconscious but breathing normally, place in recovery position and seek medical advice. Do not leave affected person unattended.

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Following inhalation:

Provide fresh air. In case of respiratory tract irritation, consult a physician.

In case of skin contact:

After contact with skin, wash immediately with plenty of water and soap. Take off immediately all contaminated clothing.
If skin irritation or rash occurs: Get medical advice/attention.

After eye contact:

Rinse cautiously with water for several minutes. If eye irritation persists: Get medical advice/attention. Remove contact lenses, if present and easy to do. Continue rinsing.

Following ingestion:

Rinse mouth. Let 1 glass of water be drunken in little sips (dilution effect). Get medical advice/attention if you feel unwell.

Self-protection of the first aider:

Use personal protection equipment. No direct artificial respiration to be given by first aider.

4.2. Most important symptoms and effects, both acute and delayed

Skin corrosion/irritation, Serious eye damage/eye irritation

4.3. Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

SECTION 5: Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media:

Full water jet, Carbon dioxide (CO₂), alcohol resistant foam

Unsuitable extinguishing media:

Full water jet

5.2. Special hazards arising from the substance or mixture

Hazardous combustion products:

In case of fire: Gases/vapours, toxic

5.3. Advice for firefighters

Wear a self-contained breathing apparatus and chemical protective clothing.

5.4. Additional information

Collect contaminated fire extinguishing water separately. Do not allow entering drains or surface water.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

6.1.1. For non-emergency personnel

Personal precautions:

Remove persons to safety.

Protective equipment:

Wear protective gloves/protective clothing/eye protection/face protection.

6.1.2. For emergency responders

Personal protection equipment:

Personal protection equipment: see section 8

6.2. Environmental precautions

Do not allow to enter into surface water or drains.

6.3. Methods and material for containment and cleaning up

For containment:

Absorb with liquid-binding material (sand, diatomaceous earth, acid- or universal binding agents).

For cleaning up:

Water (with cleaning agent)

6.4. Reference to other sections

Safe handling: see section 7, Personal protection equipment: see section 8, Disposal: see section 13

6.5. Additional information

Use appropriate container to avoid environmental contamination.

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SECTION 7: Handling and storage

7.1. Precautions for safe handling

Protective measures

Advices on safe handling:

Wear personal protection equipment (refer to section 8).

Advices on general occupational hygiene

When using do not eat, drink, smoke, sniff. Avoid contact with skin, eyes and clothes.

7.2. Conditions for safe storage, including any incompatibilities

Technical measures and storage conditions:

Keep container tightly closed in a cool, well-ventilated place.

Requirements for storage rooms and vessels:

Recommended storage temperature: at room temperature.

Storage class (TRGS 510, Germany): 12 – non-combustible liquids that cannot be assigned to any of the above storage classes

7.3. Specific end use(s)

No data available

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

No data available

8.2. Exposure controls

8.2.1. Appropriate engineering controls

No data available

8.2.2. Personal protection equipment



Eye/face protection:

Eye glasses with side protection EN 166

Skin protection:

Tested protective gloves must be worn (EN ISO 374). Take recovery periods for skin regeneration. Breakthrough times and swelling properties of the material must be taken into consideration. Suitable material: NBR (Nitrile rubber), Thickness of the glove material: > 0,35 mm. In the case of wanting to use the gloves again, clean them before taking off and air them well.

Respiratory protection:

Usually no personal respiratory protection necessary. In case of inadequate ventilation wear respiratory protection. Filtering device with filter or ventilator filtering device of type: ABEK-P3.

Other protection measures:

Wash hands before breaks and after work.

8.2.3. Environmental exposure controls

No data available

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance

Physical state: Liquid

Form: Liquid

Colour: colourless

Odour: not determined

flammability: No

Safety relevant basis data

Parameter	Value	at °C	① Method ② Remark
pH	No data available		
Melting point	No data available		

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Parameter	Value	at °C	① Method ② Remark
Freezing point	No data available		
Initial boiling point and boiling range	100 °C		
Flash point	No data available		
Evaporation rate	No data available		
Auto-ignition temperature	not applicable		
Upper/lower flammability or explosive limits	No data available		
Vapour pressure	No data available		
Vapour density	No data available		
Density	1.133 g/cm³	20 °C	
Bulk density	not applicable		
Water solubility	completely miscible		
Dynamic viscosity	No data available		
Kinematic viscosity	No data available		

9.2. Other information

No data available

SECTION 10: Stability and reactivity

10.1. Reactivity

No data available

10.2. Chemical stability

No data available

10.3. Possibility of hazardous reactions

No data available

10.4. Conditions to avoid

No data available

10.5. Incompatible materials

No data available

10.6. Hazardous decomposition products

No data available

SECTION 11: Toxicological information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

guanidinium chloride CAS No.: 50-01-1 EC No.: 200-002-3
LD₅₀ oral: 729.8mg/kg (rat) other: Health effects test guidelines, August 1982, EPA 560/6-82-001
LD₅₀ dermal: >2,000mg/kg (rabbit) other: EPA TS-792 Acute exposure, dermal toxicity. Health effects test guidelines, EPA, August 1982; EPA 560/6-82-001
LC₅₀ Acute inhalation toxicity (dust/mist): 3.181mg/L 4h (Rat)
cetrimonium bromide CAS No.: 57-09-0 EC No.: 200-311-3
LD₅₀ oral: 410mg/kg (Rat)

Acute oral toxicity:

Harmful if swallowed.

Acute dermal toxicity:

Based on available data, the classification criteria are not met.

Acute inhalation toxicity:

Based on available data, the classification criteria are not met.

Skin corrosion/irritation:

Causes skin irritation.

Serious eye damage/irritation:

Causes serious eye irritation.

Respiratory or skin sensitisation:

Based on available data, the classification criteria are not met.

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Germ cell mutagenicity:

Based on available data, the classification criteria are not met.

Carcinogenicity:

Based on available data, the classification criteria are not met.

Reproductive toxicity:

Based on available data, the classification criteria are not met.

STOT-single exposure:

Based on available data, the classification criteria are not met.

STOT-repeated exposure:

Based on available data, the classification criteria are not met.

Aspiration hazard:

Based on available data, the classification criteria are not met.

Additional information:

No data available

11.2. Information on other hazards

No data available

SECTION 12: Ecological information

12.1. Toxicity

guanidinium chloride CAS No.: 50-01-1 EC No.: 200-002-3
LC₅₀: 2.420mg/L 2d (fish, <i>Leuciscus idus</i>) other: DIN 38412 L15 (DEV-11. Lieferung 1982)
EC₅₀: 33.5mg/L 3d (Algae/water plant, <i>Raphidocelis subcapitata</i> (previous names: <i>Pseudokirchneriella subcapitata</i> , <i>Selenastrum capricornutum</i>)) EU Method C.3 (Algal Inhibition test)
EC₅₀: 70.2mg/L 2d (crustaceans, <i>Daphnia magna</i>) OECD Guideline 202 (<i>Daphnia</i> sp. Acute Immobilisation Test)
NOEC: 6.3mg/L 3d (Algae/water plant, <i>Raphidocelis subcapitata</i> (previous names: <i>Pseudokirchneriella subcapitata</i> , <i>Selenastrum capricornutum</i>)) EU Method C.3 (Algal Inhibition test)
NOEC: ≥181mg/L 35d (fish, <i>Pimephales promelas</i>) OECD Guideline 210 (Fish, Early-Life Stage Toxicity Test)
NOEC: 2.9mg/L 21d (crustaceans, <i>Daphnia magna</i>) OECD Guideline 211 (<i>Daphnia magna</i> Reproduction Test)
LOEC: ≥424mg/L 35d (fish, <i>Pimephales promelas</i>) OECD Guideline 210 (Fish, Early-Life Stage Toxicity Test)
LOEC: 4.2mg/L 21d (crustaceans, <i>Daphnia magna</i>) OECD Guideline 211 (<i>Daphnia magna</i> Reproduction Test)
cetrionium bromide CAS No.: 57-09-0 EC No.: 200-311-3
LC₅₀: 0.2mg/L 4d (fish, <i>Danio rerio</i> (previous name: <i>Brachydanio rerio</i>))
EC₅₀: 0.00411mg/L 3d (Algae/water plant, <i>Raphidocelis subcapitata</i> (previous names: <i>Pseudokirchneriella subcapitata</i> , <i>Selenastrum capricornutum</i>))
EC₅₀: 0.026mg/L 2d (crustaceans, <i>Daphnia magna</i>)
NOEC: 0.0011mg/L 3d (Algae/water plant, <i>Raphidocelis subcapitata</i> (previous names: <i>Pseudokirchneriella subcapitata</i> , <i>Selenastrum capricornutum</i>))
NOEC: 0.023mg/L 21d (crustaceans, <i>Daphnia magna</i>) OECD Guideline 211 (<i>Daphnia magna</i> Reproduction Test)

Aquatic toxicity:

Very toxic to aquatic life. Harmful to aquatic life with long lasting effects.

12.2. Persistence and degradability

No data available

12.3. Bioaccumulative potential

guanidinium chloride CAS No.: 50-01-1 EC No.: 200-002-3
Log K_{OW}: < 0.02
Bioconcentration factor (BCF): 0.893
cetrionium bromide CAS No.: 57-09-0 EC No.: 200-311-3
Log K_{OW}: 3.18
Bioconcentration factor (BCF): 80.4 Species: <i>Lepomis macrochirus</i>

12.4. Mobility in soil

No data available

12.5. Results of PBT and vPvB assessment

guanidinium chloride CAS No.: 50-01-1 EC No.: 200-002-3
Results of PBT and vPvB assessment: —

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cetrimonium bromide CAS No.: 57-09-0 EC No.: 200-311-3

Results of PBT and vPvB assessment: This substance does not meet the PBT/vPvB criteria of REACH, Annex XIII.

12.6. Endocrine disrupting properties

No data available

12.7. Other adverse effects

No data available

SECTION 13: Disposal considerations

13.1. Waste treatment methods

Waste treatment options

Appropriate disposal / Product:

Consult the appropriate local waste disposal expert about waste disposal.

SECTION 14: Transport information

Land transport (ADR/RID)	Inland waterway craft (ADN)	Sea transport (IMDG)	Air transport (ICAO-TI / IATA-DGR)
14.1. UN number or ID number			
UN 3082	UN 3082	UN 3082	UN 3082
14.2. UN proper shipping name			
ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (cetrimonium bromide)	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (cetrimonium bromide)	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (cetrimonium bromide)	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (cetrimonium bromide)
14.3. Transport hazard class(es)			
 9	 9	 9	 9
14.4. Packing group			
III	III	III	III
14.5. Environmental hazards			
		 MARINE POLLUTANT	
14.6. Special precautions for user			
Special Provisions: 274 335 375 601 650 Limited quantity (LQ): 5 L Excepted Quantities (EQ): E1 Hazard identification number (Kemler No.): 90 Classification code: M6 Tunnel restriction code: (-)	Special Provisions: 274 335 375 601 650 Limited quantity (LQ): 5 L Excepted Quantities (EQ): E1 Classification code: M6	Special Provisions: 274 335 375 969 Limited quantity (LQ): 5 L Excepted Quantities (EQ): E1 EmS-No.: F-A, S-F	Special Provisions: A97 A158 A197 A215 Limited quantity (LQ): Y964 Excepted Quantities (EQ): E1

14.7. Maritime transport in bulk according to IMO instruments

No data available

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.1.1. EU legislation

Other regulations (EU):

Directive 2012/18/EU on the control of major-accident hazards involving dangerous substances [Seveso-III-Directive],

Hazard categories:

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- E1 Hazardous to the Aquatic Environment in Category Acute 1 or Chronic 1
1907/2006 REACH, 1272/2008 CLP GHS, 98/24/EC

15.1.2. National regulations

[DE] National regulations

Restrictions of occupation

22 JArbSchG., 4 MuSchRiV., 5 MuSchRiV., Betriebssicherheitsverordnung (BetrSichV)

Störfallverordnung (12. BlmschV)

for substances contained in the product:

Hazard categories:

- E1 Hazardous to the Aquatic Environment in Category Acute 1 or Chronic 1

Water hazard class

WGK:

3 - highly hazardous to water

Berufsgenossenschaftliche Vorschriften (DGUV-Vorschriften)

Berufsgenossenschaftliche Vorschriften (DGUV-Vorschriften)

Other regulations, restrictions and prohibition regulations

Hazardous Substances Ordinance (GefStoffV)

15.2. Chemical Safety Assessment

No data available

SECTION 16: Other information

16.1. Indication of changes

No data available

16.2. Abbreviations and acronyms

ADN	European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
BCF	Bioconcentration Factor
CAS	Chemical Abstracts Service
CLP	Classification, Labelling and Packaging
DIN	German Institute for Standardization / German Industrial Standard
EC ₅₀	Effective Concentration 50%
ECHA	European Chemicals Agency
EN	European Standard
ICAO	International Civil Aviation Organization
IMDG	International Maritime Dangerous Goods
IMO	International Maritime Organization
ISO	International Standards Organisation
LC ₅₀	Lethal (fatal) Concentration 50%
LD ₅₀	Lethal (fatal) Dose 50%
NFPA	National Fire Protection Association
NOEC	No Observed Effect Concentration
OECD	Organisation for Economic Cooperation and Development
PBT	persistent and bioaccumulative and toxic
REACH	Registration, Evaluation and Authorization of Chemicals
RID	Dangerous goods regulations for transport by rail
SCL	Specific concentration limit
TRGS	Technische Regeln für Gefahrstoffe
UN	United Nations

For abbreviations and acronyms, see: ECHA Guidance on information requirements and chemical safety assessment, chapter R.20 (Table of terms and abbreviations).

16.3. Key literature references and sources for data

Safety data sheets for the ingredients, 1907/2006 REACH, 1272/2008 CLP GHS, TRGS 525, TRGS 510, TRGS 900

Substance name	Type	source of supply
guanidinium chloride CAS No.: 50-01-1 EC No.: 200-002-3	LD ₅₀ oral; LD ₅₀ dermal; LC ₅₀ ; EC ₅₀ ; NOEC; LOEC	Source: European Chemicals Agency, http://echa.europa.eu/
cetrimonium bromide CAS No.: 57-09-0 EC No.: 200-311-3	LC ₅₀ ; EC ₅₀ ; NOEC	Source: European Chemicals Agency, http://echa.europa.eu/

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16.4. Classification for mixtures and used evaluation method according to regulation (EC) No 1272/2008 [CLP]

Hazard classes and hazard categories	Hazard statements	Classification procedure
Acute toxicity (oral) (<i>Acute Tox. 4</i>)	H302: Harmful if swallowed.	Calculation method.
Skin corrosion/irritation (<i>Skin Irrit. 2</i>)	H315: Causes skin irritation.	Calculation method.
Serious eye damage/eye irritation (<i>Eye Irrit. 2</i>)	H319: Causes serious eye irritation.	Calculation method.
Hazardous to the aquatic environment (<i>Aquatic Acute 1</i>)	H400: Very toxic to aquatic life.	Calculation method.
Hazardous to the aquatic environment (<i>Aquatic Chronic 3</i>)	H412: Harmful to aquatic life with long lasting effects.	Calculation method.

16.5. List of relevant hazard statements and/or precautionary statements from sections 2 to 15

Hazard statements	
H302	Harmful if swallowed.
H314	Causes severe skin burns and eye damage.
H315	Causes skin irritation.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H335	May cause respiratory irritation.
H373	May cause damage to organs through prolonged or repeated exposure.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

16.6. Training advice

No data available

16.7. Additional information

This data sheet was prepared in accordance with EU Regulation (EC) No. 1907/2006 (REACH) and, where applicable, US Regulation 29 CFR 1910.1200, Appendix D.

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